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PAPER

YOLOv10n-BC: A Novel Real-Time Object Detection Model for Driver Distracted Driving Detection

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SUMMARY Driver distraction is a primary cause of traffic accidents, and the real-time and effective detection of such behaviors can significantly reduce traffic-related injuries and fatalities. In this paper, we enhance the lightweight YOLOv10n model by integrating the BiFPN structure to bolster its multi-scale feature extraction capabilities. Additionally, we design a CASSA module that combines channel attention, spatial attention, and channel shuffle to strengthen the model's ability to capture long-range dependencies. The model was tested on the CBTDDD dataset, established in this study, which includes data on driver distraction across multiple scenarios involving sedans, passenger buses, and trucks. Compared to the original YOLOv10n model, the proposed model demonstrates a 2.0% improvement in mAP@0.5 and achieves an FPS of 115.3 f/s. These results indicate that the YOLOv10n-BC model developed in this paper is capable of performing real-time and efficient monitoring of driver distraction.

key words: *distracted driving detection, YOLOv10n, multi-scale features, channel shuffle, BiFPN*

1. Introduction

With the robust growth of the global economy and the continuous expansion of the transportation industry, the incidence of road traffic accidents has been increasing year by year, becoming one of the leading causes of death worldwide. According to data from the World Health Organization (WHO), approximately 1.35 million people die in traffic accidents annually, with 50 million injured, making traffic accidents the second leading health threat after diseases [1]. Statistics from the National Highway Traffic Safety Administration (NHTSA) indicate that distracted driving is a major cause of severe traffic accidents, with 3,308 fatalities in the United States in 2022 alone due to motor vehicle collisions resulting from distracted driving [2]. These accidents often occur when drivers engage in activities that divert their attention from the road, such as making phone calls, texting, eating, conversing with passengers, and manipulating audio, entertainment, or navigation systems. To minimize the occurrence of traffic accidents, motor vehicle safety and the assurance of safe driving have become research hotspots for experts both domestically and internationally.

Although autonomous driving technology envisions a future where driving may not require driver participation at all, the technology is still in its developmental stages. Factors such as technological immaturity, incomplete regulations, and societal acceptance suggest that widespread application is still a long way off. Therefore, during this transitional period, researching and developing effective distracted driving detection technologies is of great significance for reducing traffic accidents and ensuring safe driving.

Distracted driving detection technologies are primarily categorized into three types based on the acquisition method of natural driving data: detection based on physiological signals, detection based on vehicle sensors, and detection based on images. Detection methods based on physiological signals analyze physiological indicators such as electroencephalograms (EEG) and electrocardiograms (ECG) to assess the driver's attention state. While accurate, these methods are costly, inconvenient to use, and somewhat invasive. Methods based on vehicle sensors focus on analyzing vehicle control data such as speed, braking distance, and steering wheel angle, which can indirectly reflect the driver's behavior patterns. However, they suffer from latency, inability to provide real-time warnings, significant environmental influence, and large variations in sensor data across different manufacturers and models [3], [4]. Image-based detection methods have garnered widespread attention due to their non-contact and easy deployment characteristics, making them particularly suitable for real-time monitoring in vehicle environments.

In the field of image-based distracted driving detection, traditional machine learning methods typically rely on manually extracted features such as facial keypoints, posture information, and eye closure rate, using classifiers like Support Vector Machines (SVM) and decision trees for distracted state recognition. These methods require substantial domain knowledge for feature extraction and classifier design and have limited performance in complex scenarios. With the advancement of computer vision technology, deep learning-based methods have gradually become a research focus. Deep learning models, especially Convolutional Neural Networks (CNNs), can automatically learn and extract effective features, significantly enhancing the accu-

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racy and robustness of distracted driving detection. Most studies focus on the driving state of the driver itself, and the research data do not consider other personnel except the driver, nor other background interference in the image, which is quite different from the actual driving monitoring scene. Therefore, the previous model lacks the research on the multi-object scene such as driver and passenger in the actual driving scene. those model's detection efficiency and real-time performance of monitoring data in real scenarios are insufficient.

In response to the aforementioned issues, this paper proposes a YOLO series-based distracted driving detection method. Specifically, this paper employs YOLOv10n as the base detection model, combined with a Bidirectional Feature Pyramid Network (BiFPN) and a Channel Attention, Spatial Attention, and Channel Shuffle Integrated Module (CASSA), to enhance the model's feature expression capability and detection accuracy. The YOLOv10n+BiFPN+CASSA (YOLOv10n-BC) model not only achieves efficient multi-object detection but also excels in real-time performance, with a detection frame rate (FPS) of 115 f/s, meeting the high real-time requirements of practical applications.

The contributions of this study are:

- This research creates a dataset named CBTDDD, covering distracted driving scenarios of drivers in four vehicle types: sedans, passenger cars, buses, and trucks. The dataset collects a diverse range of driving images, providing a rich set of samples for training and evaluating distracted driving detection models.
- We have developed the CASSA module, integrating channel attention, channel shuffle, and spatial attention mechanisms. This module effectively captures global dependencies in feature maps, enhances feature representation capabilities, and improves detection accuracy.
- Based on YOLOv10n, combined with BiFPN and the CASSA module, we have constructed a lightweight distracted driving detection model. This model not only improves detection rates but also optimizes computational efficiency, making it suitable for real-time monitoring applications.

2. Related Work

Image-based driver distraction detection technology, as an important means of improving road traffic safety, has received widespread research attention in recent years. With the rapid development of deep learning technology, many efficient methods have been proposed to improve the accuracy and real-time performance of detection. Early research mainly relied on Convolutional Neural Networks (CNNs) to automatically extract driver behavior features, demonstrating excellent performance in image classification tasks. For instance, Chao Yan et al. [5] used CNNs to directly learn discriminative features from raw images, classifying driving postures into four categories: normal driving, gear shifting, eating and smoking, and answering mobile phones, achieving an overall accuracy of 99.78% on the Southeast University Driving Posture Dataset. Additionally, Bhakti [6] employed a depth-separable convolution-based MobileVGG network, effectively compressing model parameters to 2.2M, and achieved accuracies of 95.24% and 99.75% on the AUC dataset and State Farm dataset, respectively, showcasing the potential of lightweight networks to maintain high performance while significantly reducing computational complexity. As research progressed, attention mechanisms were introduced into distracted driving detection models to enhance the model's ability to focus on key features. Wang et al. [7] incorporated a Channel and Space Attention Module (CSAM) into VGG16 and ResNet50 models, and experimental results on the SFDDD dataset showed improved recognition accuracy with the attention mechanism. Imen Jegham et al. [8] proposed a novel Depth-based Spatial Attention network (DSA) that integrates depth information into RGB data, combined with a pre-trained VGG16 model and LSTM network, for driver behavior recognition tasks on the MDAD dataset, achieving up to a 27% increase in accuracy compared to existing technologies. Huang et al. [9] designed a detection method based on the driver's true focus of attention (TDFoA), combining deep 3D residual networks with attention mechanisms, enabling precise detection of distracted states without the need for threshold setting, significantly enhancing recognition accuracy across multiple datasets. These studies indicate that attention mechanisms not only improve the model's recognition capabilities but also effectively filter out irrelevant information in complex scenarios, enhancing the model's robustness.

To further enhance detection performance, multimodal data fusion methods have also been widely applied in distracted driving detection. Yehya Abouelnaga et al. [10] created the AUC dataset and proposed a genetically weighted ensemble CNN system, utilizing raw images, facial images, and hand images for training, and optimizing the weighted combination of different model outputs to achieve high-precision classification. Hesham M. Eraqi et al. [11] also used genetically weighted ensemble CNNs, combined with multivariate Gaussian Naive Bayes classifiers for skin segmentation, and utilized pre-trained models for facial and hand detection, further improving the accuracy of distracted recognition. Furkan Omerustaoglu et al. [12] proposed a two-stage system, first constructing a visual CNN model through transfer learning and fine-tuning techniques, then integrating sensor and image data with LSTM-RNN models, enhancing the detection capability for distracted behaviors. These multimodal fusion methods, by integrating different types of data sources, provide more comprehensive driver behavior information, thereby significantly enhancing the overall performance of the detection system.

Temporal modeling is also an essential means of capturing the dynamic changes in driver behavior. Manuel Martin et al. [13] proposed a three-stream RNN architecture that includes temporal, spatial, and contextual streams, performing action recognition based on skeleton and contextual infor-

mation, better capturing the temporal dynamics and spatial states of actions. However, such models face challenges of high computational complexity and insufficient generalization ability when dealing with long-duration tasks. Patrick Gebert et al. [14] combined 3D CNNs with LSTM networks, automatically learning features from video data by analyzing the driver's head movements and body language. Although feature design is not required, the application on small-scale datasets may limit the model's generalizability. Jimiama Mafeni Mase et al. [15] proposed a hybrid model combining CNNs and stacked bidirectional LSTMs, achieving a 92.7% accuracy rate on the AUC dataset, although there are certain limitations in detecting unseen distracted driving postures. These temporal models, by capturing the temporal information of actions, enhance the recognition capability for dynamic distracted behaviors. However, a balance between model complexity and real-time performance is still needed in practical applications.

Feature extraction and image preprocessing technologies also play a crucial role in distracted driving detection. Maitree Leekhaa [16] used the GrabCut algorithm for preprocessing images obtained from vehicle cameras, combining features of the driver's posture, head, arms, and body posture, achieving accuracies of 98.48% and 95.64% on the State Farm Distracted Driver Detection dataset (SFD3) and AUC Distracted Driver dataset (AUCD2), respectively. Yang et al. [17] used a Gaussian Mixture Model (GMM) for segmenting raw RGB images, extracting the driver area, and then adjusting and training AlexNet, GoogLeNet, and ResNet50 to recognize normal driving and six types of distracted behaviors. Binbin Qin et al. [18] proposed the D-HCNN model, combining HOG features with a CNN structure that reduces filter size, improving performance through L2 regularization, Dropout, and batch normalization, achieving efficient real-time distracted driving detection. Effective feature extraction and preprocessing not only enhance the model's detection accuracy but also reduce noise interference and enhance the system's robustness.

In addition to the aforementioned main methods, technologies based on transformers have also been explored in recent years to enhance the performance of distracted driving detection. For example, Junwei Liang et al. [19] proposed a transformer-based Stargazer system, learning spatiotemporal features through a multi-scale visual transformer network and using a sliding window classification strategy, performing excellently in the AI City Challenge 2022. Mohammed et al. [20] (2024) proposed an innovative semi-supervised learning method using a lightweight vision transformer to improve the detection accuracy of driver distraction behaviors. This method effectively uses unlabeled data by combining pseudo-label generation and semi-supervised learning strategies, reducing the dependence on a large amount of labeled data. In addition, researchers have used a hybrid lightweight transformer model trained in a teacher-student network framework to achieve small model size and fast inference. Mou et al. [21] (2023) proposed a dual-channel network combining CNNs and transformers for multimodal driver distraction detection. The network optimizes feature extraction through midpoint residual structures and uses double-layer optimization to automatically adjust hyperparameters. Experimental results show that this method is effective in improving detection accuracy. These advanced methods, by introducing transformer models and integrating various features, further enhance the accuracy and robustness of distracted driving detection. However, the high computational and memory demands of transformer models pose challenges for real-time applications.

In terms of dataset construction and utilization, in addition to a few researchers using their own privately created datasets, most researchers' studies mainly focus on public datasets. The most commonly used datasets are three: the State Farm Distracted Driver Detection dataset (SFDDD) first appeared in [22], annotating driver driving image data into ten categories: safe driving, texting with the right hand, talking on the phone with the right hand, texting with the left hand, talking on the phone with the left hand, operating the radio, drinking water, reaching backward, grooming hair or applying makeup, and talking with passengers. The AUC Distracted Driver Monitoring Dataset is also a commonly used dataset, first appearing in [10]. This dataset marks the driving situation of drivers recorded by a monocular camera, similar to the State Farm Distracted Driver Detection dataset, dividing driving behaviors into the same ten categories. This dataset is suitable for both image models and video models. The Distracted Driver Monitoring Dataset (DMD) was first constructed by Ortega et al. [23], and the DMD dataset includes videos of drivers' faces, hands, and bodies recorded from three in-vehicle perspectives, with each camera providing RGB, infrared, and depth information. The dataset includes data from 37 volunteers, 27% of whom are women. The DMD dataset uses a temporal annotation mechanism, with each video sequence (frame group) marked as one category. Additionally, some studies have pointed out that there are certain annotation errors in the AUC dataset and DMD dataset [24]. To be compatible with both image and video models, this paper created a distracted driving detection image dataset CBTDDD in this work, covering distracted driving behaviors of drivers in sedans, passenger cars, buses, and trucks, aiming to cover driver behavior characteristics under different vehicle types and further enhance the model's generalization ability and application range.

In summary, the field of driver distraction detection has made many advancements in deep learning model optimization, application of attention mechanisms, multimodal data fusion, temporal modeling, and feature extraction and image preprocessing. Existing research has significantly improved the accuracy and robustness of detection by using convolutional neural networks to automatically extract efficient features, combining attention mechanisms to focus on key areas, and integrating multimodal data and temporal sequence information. However, existing methods still face certain challenges in terms of adaptability to different vehicle types, real-time detection efficiency, and recognition capabilities for complex distracted behaviors. Deeper models often have

high computational complexity, making it difficult to achieve low-latency operation in real-time systems; multimodal and temporal sequence methods, although enhancing recognition performance, face issues with data synchronization, alignment, and high computational resource demands in practical applications. In addition, advanced architectures such as transformers and hybrid models have shown promising results but require further optimization to balance performance and computational efficiency.

Based on the aforementioned analysis, in the preliminary work of this paper, we created a more representative dataset named CBTDDD, which covers distracted driving behaviors of drivers in passenger cars, buses, and trucks. This aims to enhance the model's generalization capabilities and applicability. Concurrently, we designed the CASSA module, which integrates channel attention, channel shuffle, and spatial attention mechanisms, to effectively capture global dependencies and strengthen feature representation capabilities. Additionally, we proposed a lightweight YOLOv10n distracted driving detection model that combines BiFPN with CASSA. This model not only improves detection accuracy but also optimizes computational efficiency, significantly enhancing real-time detection performance.

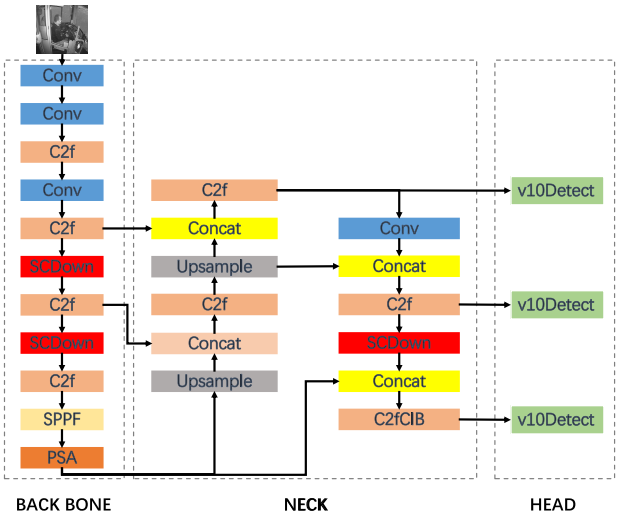
3. Methodology

3.1 Overall Architecture of YOLOv10

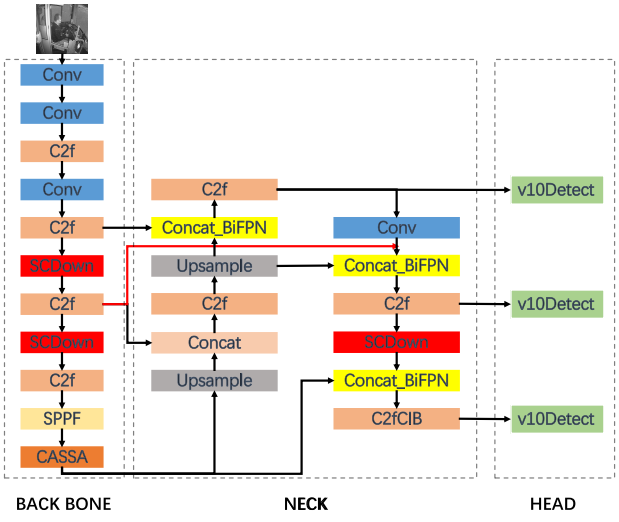
The model in this paper has modified the YOLOv10n [25] model. The YOLOv10n network is composed of three main modules: the Backbone, the Neck, and the Head, as shown in Fig. 1 (a). The Backbone module is responsible for extracting features from the input image. The Neck module processes these features to obtain three different scales of feature information. Finally, the Head module leverages these varying scales of features for detection and outputs the final results. To address issues such as significant scale variation of drivers in images and the original model's insufficient capture of global feature information, we have made certain improvements to YOLOv10n in this study, as depicted in Fig. 1 (b). These improvements are detailed as follows:

In the Neck section of the network, we have implemented the BiFPN (Bidirectional Feature Pyramid Network) structure, which enables multi-scale feature map fusion with minimal information loss. BiFPN facilitates efficient transfer and fusion of information between feature maps of different scales through bidirectional connections, thereby enhancing the model's detection capabilities for targets of varying sizes and improving the representational power of feature maps. This design not only reduces information loss during feature map fusion but also optimizes the model's inference speed and accuracy.

To effectively capture global dependencies within feature maps, we have designed the CASSA module to replace the original PSA. This module integrates channel attention, channel shuffle, and spatial attention mechanisms. Channel



(a)



(b)

Fig. 1 Comparison between YOLOv10n and YOLOv10n-BC model structure (a) YOLOv10n model structure; (b) YOLOv10n-BC model structure

attention and channel shuffle fully harness inter-channel relationships, while the spatial attention mechanism efficiently utilizes spatial information. A comparison of the YOLOv10n and YOLOv10n-BC model architectures is shown in Fig. 1.

3.2 Multi-Scale Feature Fusion with the BiFPN Module

In this study, we have enhanced the neck section of YOLOv10n by employing the BiFPN structure to better integrate multi-scale feature information. BiFPN is an improvement over the FPN and PANet architectures, aiming to reduce information loss during the fusion of multi-scale feature information.

3.2.1 FPN (Feature Pyramid Network)

Figure 2 (a) below illustrates the structure of FPN [26]. FPN

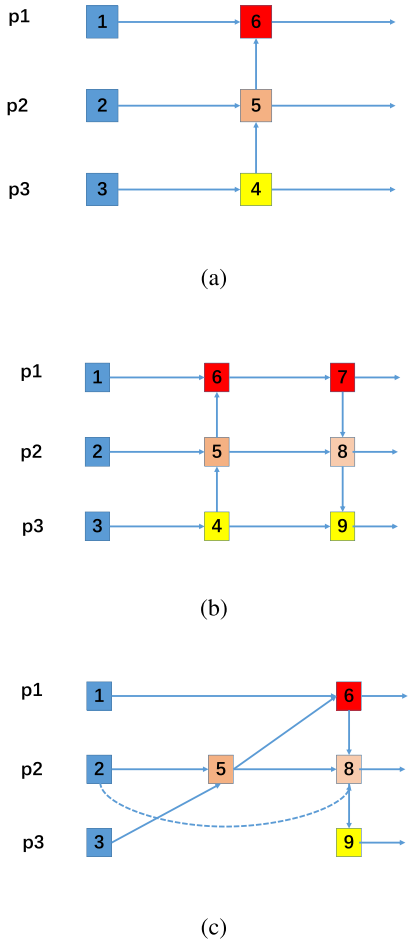


Fig. 2 Multi-scale module structure comparison: (a) FPN structure; (b) PANet structure; (c) BiFPN structure

is a classic feature pyramid network that conveys high-level semantic features to lower-level feature maps through a top-down pathway and lateral connections. Feature maps are upsampled from the top layer and fused with corresponding lower-level feature maps. Although FPN has achieved certain effects in multi-scale feature fusion, the information transfer solely through the top-down pathway may be insufficient, limiting the expressive power of features.

3.2.2 PANet

As shown in Fig. 2 (b), PANet [27] adds a bottom-up pathway to the FPN, enhancing multi-scale feature fusion through bidirectional information transfer. This structure not only transfers information from the top layer to the bottom layer but also facilitates information transfer from the bottom layer to the top layer. While PANet achieves better feature fusion by adding pathways, there is still some redundancy and information loss in the information transfer process.

3.2.3 BiFPN

As shown in Fig. 2 (c), BiFPN [28] similarly adopts bidirectional information flow mechanisms encompassing both top-down and bottom-up pathways. Compared to PANet, BiFPN demonstrates superior performance in feature fusion and object detection accuracy.

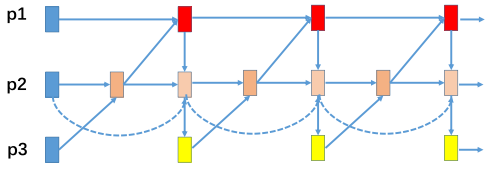


Fig. 3 BiFPN layers stacking example

The BiFPN excels in integrating high-level semantic information with low-level detailed features. This capability stems from its additional cross-layer connections, which enable direct fusion of advanced features with elementary ones, thereby harmonizing high-level semantics and low-level spatial details to enhance detection accuracy. As depicted in Fig. 2 (c), the dashed connection spanning from Node 2 to Node 8 exemplifies these cross-layer linkages. Such architectural refinements facilitate explicit interaction between abstract semantic representations from deeper layers and localized textural patterns from shallower hierarchies. In addition, BiFPN layers can be stacked repeatedly to achieve multi-stage feature interactions across layers, forming a denser information flow network. Figure 3 illustrates a configuration in which duplicate blocks are stacked three times and EfficientDet uses six duplicates. In order to realize real-time monitoring, this paper controls the number of parameters of the module, and does not use the repeated stacking capability of BiFPN.

BiFPN preserves the representational capacity of feature maps while mitigating unnecessary information loss and redundancy. It posits that single-input nodes contribute limited benefits to feature fusion, whereas redundant structures increase parametric complexity and computational workload. As illustrated in Fig. 2 (b), PANet contains a single-input node (Node 4), characterized by only one input source, along with a redundant node (Node 7) whose information duplicates that of Node 6. BiFPN addresses these inefficiencies by eliminating both the single-input Node 4 and redundant Node 7 without compromising detection accuracy.

Furthermore, BiFPN autonomously optimizes the fusion process across multi-scale feature maps. In PANet, feature fusion typically relies on naive summation or concatenation operations, which indiscriminately treat all features without prioritizing their relative importance. In contrast, BiFPN introduces learnable weight parameters to dynamically recalibrate feature contributions during fusion. This adaptive weighting mechanism, formalized in Eq. (1), enables the network to emphasize discriminative features while suppressing less informative components, thereby enhancing both efficiency and precision in multi-scale representation learning.

$$P_i^{\text{out}} = \frac{\sum_j w_j \cdot P_j^{\text{in}}}{\sum_j w_j + \epsilon} \tag{1}$$

In the equation, P_i^{out} represents the fused feature map,

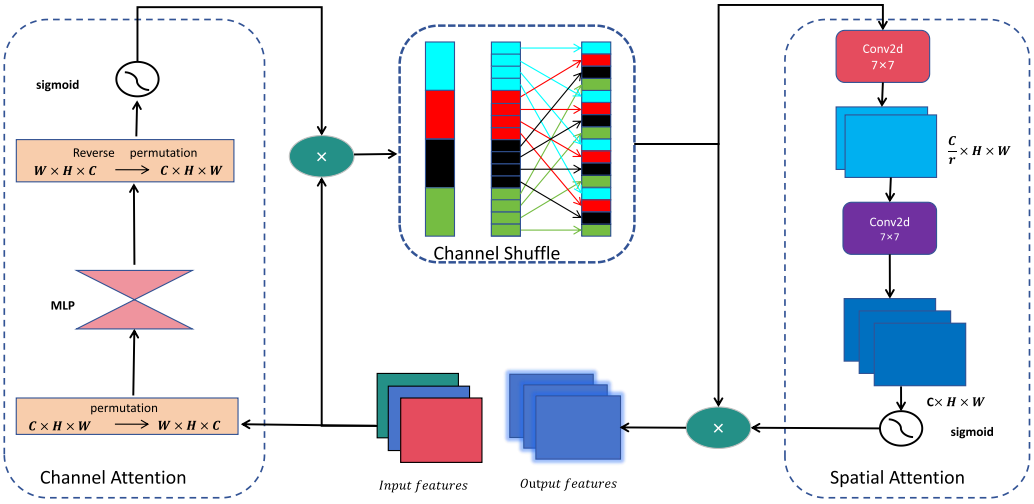


Fig. 4 CASSA module structure diagram

P_j^{in} denotes the input feature maps, w_j refers to the weights assigned to each feature map, and is a small value added to prevent division by zero. Thus, in PANet (Fig. 2 (b)), the output feature of Node 6 is derived from the element-wise summation of features from Nodes 1 and 5, expressed as $P_1^{\text{in}} + P_5^{\text{in}}$ In contrast, BiFPN (Fig. 2 (c)) computes the feature of Node 6 by adaptively fusing the output features of input Nodes 1 and 5 with their corresponding learnable weights, as specified in Equation $\frac{w_1 P_1^{\text{in}} + w_5 P_5^{\text{in}}}{w_1 + w_5 + \epsilon}$, where and are normalized weights optimized during training. This approach augments the influence of critical features while suppressing less relevant ones, thereby improving detection rates. For instance, in small object detection scenarios, the weights assigned to low-level detailed features (e.g., edges or textures) may be amplified, whereas the weights of high-level semantic features (e.g., object categories) might be attenuated. Such dynamic reweighting enables adaptive and optimized feature fusion tailored to specific detection tasks, effectively balancing semantic abstraction and spatial precision without introducing parametric redundancy or excessive computational complexity.

In the neck section of YOLOv10n-BC, we have employed the ConcatBiFPN module in place of the traditional concat module, achieving a weighted fusion of feature maps. Additionally, building upon YOLOv10n and as indicated by the red lines in Fig. 1 (b), we have integrated the BiFPN structure, thereby enabling more efficient feature map fusion in the neck section.

3.3 CASSA Module

In traditional convolutional neural networks, the relationships between channels may be overlooked; however, these relationships are crucial for capturing global information. Without considering inter-channel dependencies, the model may fail to fully utilize all the information in the feature maps, leading to an insufficient grasp of global features. To enhance the expressive power of input feature maps and capture global dependencies within them, we designed a CASSA Module that integrates channel attention, channel shuffle, and spatial attention mechanisms. The specific structure and process are shown in Fig. 4 below.

Within this module, the input feature maps are sequentially processed by a Channel Attention Submodule, a Channel Shuffle Module, and a Spatial Attention Module. The dimensions of both the initial and output feature maps are maintained at $C \times H \times W$.

3.3.1 Channel Attention Submodule

In the channel attention submodule, the input feature maps are initially subjected to a dimension permutation, transforming them from the original format of $C \times H \times W$ to $W \times H \times C$. This rearrangement of data dimensions is intended to facilitate subsequent processing by the multilayer perceptron (MLP) to more effectively address dependencies between channels. Subsequently, the feature maps are processed through two layers of MLPs designed to capture channel interdependencies. The first MLP reduces the number of channels to a quarter of the original $\frac{C}{4}$, followed by the application of a ReLU activation function to introduce non-linearity and enhance the model's expressive power. Then, the second MLP restores the number of channels to the original C dimension, reconstructing the complete channel information. After MLP processing, the feature maps undergo an inverse permutation operation to revert their dimensions back to the $C \times H \times W$ format. Thereafter, a sigmoid activation function is applied to generate a channel attention map, which represents the importance weights of each channel. Finally, the input feature maps are element-wise multiplied with the generated channel attention map to obtain the enhanced feature maps. The entire process can be represented as Eq. (2).

$$F_{\text{channel}} = \sigma\left(\text{MLP}\left(\text{Permute}\left(F_{\text{input}}\right)\right)\right) \odot F_{\text{input}} \tag{2}$$

Here, F_{channel} is the enhanced feature graph, σ is the

Sigmoid function, and $\odot$ represents the multiplication-by-element, F_{input} is the original input feature graph.

3.3.2 Channel Shuffle

To further mix and share information, a channel shuffle operation is applied. The enhanced feature maps are partitioned into 4 groups, each comprising $\frac{C}{4}$ channels. A permutation operation is then performed on the grouped feature maps to shuffle the channel order within each group. Thereafter, the shuffled feature maps are reshaped back to their original format of $C \times H \times W$. This approach effectively mixes feature information and enhances the feature representation capability, as illustrated in Eq. (3).

$$F_{\text{shuffle}} = \text{ChannelShuffle}\left(F_{\text{channel}}\right) \tag{3}$$

Here, F_{shuffle} represents the shuffled feature map, and F_{channel} denotes the number of channels in the input feature map.

3.3.3 Spatial Attention Submodule

In the spatial attention sub-module, a 7×7 convolutional layer is first applied to the input feature map to reduce the number of channels to $\frac{C}{r}$ of the original one, Taking into account the real-time performance of the model and based on the experimental verification, the value of r set in this paper is c. This is followed by batch normalization and a ReLU activation function, which achieve a nonlinear transformation of the features. Thereafter, a second 7×7 convolutional layer restores the number of channels to the original, and batch normalization is applied once more. Finally, a sigmoid activation function is used to generate a spatial attention map. The channel-shuffled feature maps are then element-wise multiplied with the generated spatial attention map to obtain the final enhanced feature maps. The entire process can be represented by the following formula:

$$F_{\text{spatial}} = \sigma\left(\text{Conv}\left(\text{BN}\left(\text{ReLU}\left(\text{Conv}\left(F_{shuffle}\right)\right)\right)\right)\right) \odot F_{shuffle} \tag{4}$$

where F_{spatial} is the feature map after spatial attention, representing the overall feature map of this module, and $F_{shuffle}$ is the feature map after Channel Shuffle processing.

The following Table 1 shows the parameter distribution of each layer of each module of CASSA when rate=c.

Table 1 The parameter distribution of the CASSA module when rate=c

Module name	Layer type name	Parameters
Channel Attention Submodule	Layer 0 (Linear)	257
	Layer 2 (Linear)	512
Spatial Attention Submodule	Layer 0 (Conv2d)	12545
	Layer 1 (BatchNorm2d)	2
	Layer 3 (Conv2d)	12800
	Layer 4 (BatchNorm2d)	512

4. Experiments and Analysis

4.1 Evaluation Metrics

The following metrics were employed in this study as evaluation criteria for the experiments: Accuracy: Accuracy represents the ratio of the number of samples correctly predicted to the total number of samples. The calculation formula is as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

Precision: Precision represents the proportion of actual positive samples among the samples predicted to be positive. The calculation formula is as follows:

$$Precesion = \frac{TP}{TP + FP} \tag{6}$$

Recall: Recall represents the proportion of positive samples that are predicted to be positive among all positive samples. The calculation formula is as follows:

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

IoU: IoU measures the overlap between the face detection model's predicted face bounding box (ABOX) and the pre-labeled face bounding box (BBOX), which is the ratio of their intersection area to their union area. The formula is as follows:

$$\text{IoU} = \frac{area(ABOX \cap BBOX)}{area(ABOX \cup BBOX)} \tag{8}$$

PRC (Precision Recall Curve): The PR curve effectively illustrates the relationship between precision and recall when the data is balanced. Samples are divided into positive and negative samples based on a certain IoU threshold; those above the threshold are positive samples, and those below are negative samples. The PR curve plots Recall on the x-axis and Precision on the y-axis. A line chart is used to display the classification effect. AP: AP stands for Average Precision and represents the precision of a certain class of objects. Its value is the area enclosed by the curve and the x-axis and y-axis. AP@0.5 represents the average precision when the threshold is set to 0.5. The calculation formula is defined as follows:

$$\text{AP@}\alpha = \int_0^1 P(\gamma)\, d\gamma \tag{9}$$

mAP (Mean Average Precision): mAP is a widely used metric in object detection tasks to measure the overall detection accuracy of a model across different categories. mAP@50 specifically refers to the average precision at an IoU threshold of 0.5.

$$mAP@50 = \frac{1}{N} \sum_{i=1}^{N} AP@0.5_i \tag{10}$$

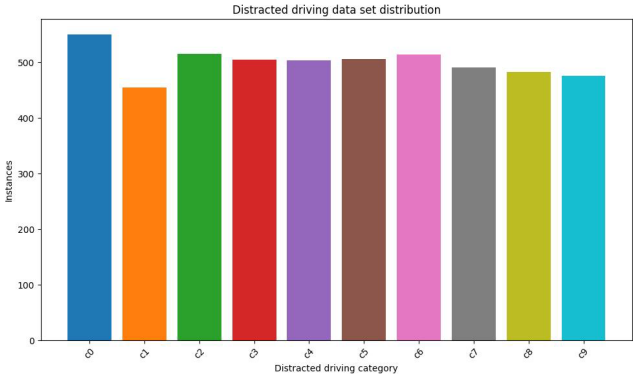


Fig. 5 Distribution of various images in CBTDDD dataset

FPS (Frames Per Second): FPS is a metric for measuring the real-time performance of a model, indicating the number of image frames the model can process per second. The higher the FPS, the faster the model's processing speed, making it suitable for real-time monitoring applications. The calculation method is as follows:

$$FPS = \frac{Number\ of\ Frames}{Time\ Taken(seconds)} \tag{11}$$

Here, *Number of Frames* is the number of image frames processed and *Time Taken*(*seconds*) is the total time required to process these frames.

4.2 Dataset Introduction

Our dataset comprises 5,000 images, with 80% allocated to the training set and 20% to the testing set. The images encompass surveillance images of sedan drivers, light passenger vehicle drivers, freight vehicle drivers, and bus drivers. Except for the sedan driver images, the images of other vehicles are sourced from the transportation management bureau's "Two Passengers and One Dangerous Goods" system, The system monitors and manages in real time the road special vehicles engaged in tourism chartered buses, three class or more line buses and transportation of hazardous chemicals, fireworks and explosives for civil use. Additionally, while the surveillance images of sedan drivers only contain the driver, other images may include other individuals, especially the bus driver surveillance images which contain a significant number of passengers. The images are categorized into 10 classes: normal driving (C0), right-hand phone use (C1), right-hand phone call (C2), left-hand phone use (C3), left-hand phone call (C4), adjusting the center console and radio (C5), drinking water (C6), right-hand reaching backward (C7), touching the body with hand distractions (C8), and conversing with passengers (C9). Examples of images for each category are shown in Fig. 5.

Images from the "Two Passengers and One Dangerous Goods" system come in various resolutions, including 352 × 288, 704 × 576, 1280 × 720, and 320 × 236 pixels. Images from sedans have a resolution of 640 × 480 pixels. Prior to training, this study uniformly converts all data to a resolution of 640 × 640 pixels. Examples of various distracted driving instances in this dataset are shown in Fig. 6.



Fig. 6 Example of 10 driving categories of CBTDDD: (a) normal driving C0; (b) using cell phone with right hand C1; (c) using cell phone with right hand; (d) using cell phone with left hand; (e) using cell phone with left hand; (f) tuning center control and radio C5; (g) drinking water C6; (h) reaching back with right hand C7; (i) hand touching body distraction C8; (j) talking to passengers C9.



Fig. 7 Demonstration of driver monitoring data labeling. (a) Car; (b) passenger car; (c) truck

Considering that the driving state is not only related to the driver themselves but may also involve devices such as the steering wheel, gear shifter, and center console, and recognizing that excessive information within the bounding box can adversely affect classification accuracy, this study standardizes the bounding box annotations during data labeling. Specifically, each bounding box precisely encompasses the driver, the center console of the driver's seat, and the gear shifter. The entire bounding box is then categorized into one of the driving states C0 through C9. Figure 7 shows the annotation examples of three typical driver monitoring images, where the green or yellow regions represent the manually marked detection boxes.

4.3 Parameter Settings

To ensure the fairness and impartiality of the experiments, this study standardized the parameters and equipment used for model training and evaluation. Specifically, Table 2 shows the settings of the training parameters, and Table 3 details the hardware and software configurations of the equipment used in both training and testing.

4.4 Experimental Process

In Fig. 8, we present a comparative analysis of the precision-recall (PR) curves for the original YOLOv10n model and the

Table 2 Model training and test parameters

Parameter	Values
batch	16
image size	640
momentum	0.937
lr	0.01
epochs	300
optimizer	SGD
final learning rate factor	0.01
weight_decay	0.0005

Table 3 Hardware and software configuration of training and testing equipment

Name	Parameter
CPU	Intel® Xeon® CPU E5-2699 v3*2
GPU	NVIDIA GeForce RTX 3090 24G
Memory	64G
OS	Ubuntu 20.04.5
Python	Python 3.9.0
PyTorch	2.0.1+cu117

enhanced YOLOv10n-BC model on the CBTDDD dataset, highlighting significant differences between them. The enhanced YOLOv10n-BC model exhibited improvements in the mean average precision (mAP@0.5) across all categories of distracted driving behaviors (c0 to c9), resulting in an overall 2% increase in the average mAP@0.5. This improvement strongly suggests that the introduced refinements effectively enhanced the model's precision in detecting various categories of driving behaviors. Specifically, the optimization of the model structure significantly improved the feature extraction capabilities, thereby enhancing the model's generalization in complex driving scenarios and ensuring robust and consistent detection performance.

Figure 9 illustrates the trend of key performance metrics of the YOLOv10n-BC model over 300 training epochs. As training progresses, box loss, object loss, and class loss exhibit a gradual decline, while metrics such as precision, mAP@0.5, mAP@0.5-0.95, and recall continue to rise. This phenomenon indicates that the YOLOv10n-BC model effectively optimizes its parameters during training, progressively learning more efficient and robust feature representations, thereby achieving a significant enhancement in object detection performance.

4.5 Ablation Study

In the ablation study, this research systematically evaluated the impact of the BiFPN and CASSA modules on the performance of the YOLOv10n model. The Fig. 10 shows that although there is a slight fluctuation in precision and recall after model convergence, the precision of models using either the BiFPN module or the CASSA module alone is higher than that of the baseline model without these modules. When both the BiFPN and CASSA modules are used together, the precision is further enhanced. Additionally, the improvements in mAP@0.5 and mAP@0.5-0.95 are even more significant, thereby further validating the effectiveness of these modules. The experimental results in the Table 4 quantify that introducing either the BiFPN module or the

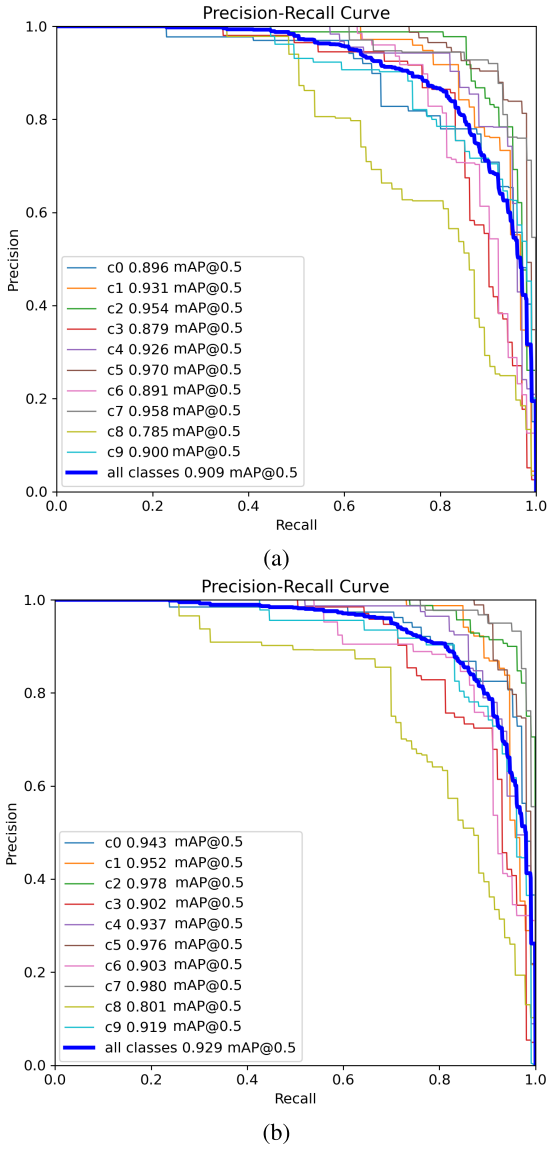


Fig. 8 PR curve of YOLOv10n and YOLOv10n-BC after training CBTDDD dataset (a) YOLOv10n (b) YOLOv10n-BC

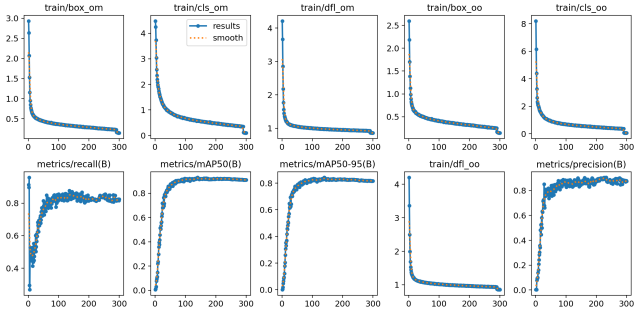


Fig. 9 Variation of box loss, the object loss, class loss, Precision, mAP50, mAP50-95 and recall'll epochs for the training datasets of the YOLOv10n-BC models.

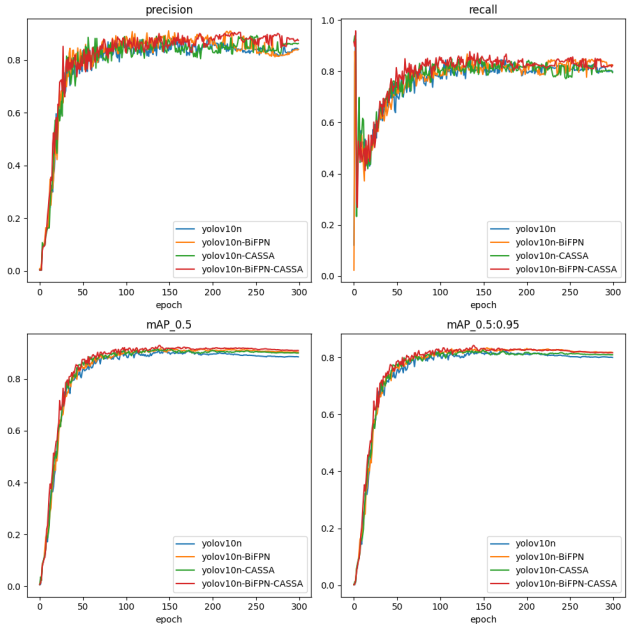


Fig. 10 Precesion, recall, mAP0.5, MAP0.5:0.9 obtained by YOLOv10n, YOLOv10n-BIFPN, YOLOv10n-CASSA and YOLOv10n-BifPn-CASSA models on the verification set during the training of 300 epochs 5 comparison of four parameters

Table 4 Comparison of ablation experiments

Method	BiFPN	CASSA	mAP50	mAP50-95	Parameter	FPS
(1) YOLOv10n+			0.909	0.825	2.698m	131.4
(2) YOLOv10n+	✓		0.915	0.834	2.715m	117.1
(3) YOLOv10n+		✓	0.919	0.826	2.477m	121.4
(4) YOLOv10n+	✓	✓	0.929	0.842	2.504m	115.3

CASSA module alone can significantly improve the model's performance on key metrics such as precision, mAP@0.5, and mAP@0.5-0.95. This indicates that on the CBTDDD dataset, the BiFPN module efficiently fuses features of different scales, enhancing the expressive power of multi-scale features and thereby strengthening the model's detection capabilities for targets of various sizes. The CASSA module effectively promotes the interaction and transmission of information between channels, thereby enhancing the model's attention to critical spatial regions. When both the BiFPN and CASSA modules are applied to the YOLOv10n model, the mAP@0.5 and mAP@0.5-0.95 metrics are improved by 2% and 1.7%, respectively, highlighting the synergistic effect of these two modules and indicating that their optimized design in feature extraction and information transmission can further enhance the model's detection performance. It is noteworthy that using either the BiFPN module or the CASSA module alone or in combination had little effect on recall. This may be because the significant improvement in precision mainly stems from the optimized model's ability to more accurately distinguish between targets and non-targets, thereby reducing false positives. However, the lack of a noticeable increase in recall may be due to the model being more conservative in its decisions, leading to some true positive samples not being detected.

Table 5 Comparison with other commonly used detection models

Method	Precision	Recall	mAP50	Parameter	FPS
YOLOv5n	0.897	0.806	0.906	**1.773m**	**205.0**
YOLOv6n	0.839	**0.841**	0.896	4.389m	83.6
YOLOv7-tiny	0.834	0.833	0.906	6.039m	202.8
YOLOv8n	0.828	0.837	0.907	3.008m	200.0
YOLOv9-tiny	0.900	0.827	0.916	2.620m	53.2
YOLOv10n	0.826	**0.841**	0.909	2.698m	131.4
YOLOv10n+CSAM	0.849	0.823	0.912	2.470m	113.4
YOLOv10n-BC	**0.903**	0.840	**0.929**	2.504m	115.3

4.6 Comparative Experiment

In this study, we conducted a comparative analysis of the performance of different versions of the YOLO series models on the CBTDDD dataset, including YOLOv5n, YOLOv6n, YOLOv7-tiny, YOLOv8n, YOLOv9-tiny, YOLOv10n, and the improved YOLOv10n+BiFPN+CASSA model. The experimental results, as indicated in Table 5, show that the improved YOLOv10n+BiFPN+CASSA model significantly outperforms the other models in terms of precision and mAP@0.5, with approximately 0.6% and 2% improvements, respectively, over the original YOLOv10n model, while maintaining a high recall rate and real-time processing capability. Although the FPS is lower than that of YOLOv5n and YOLOv7-tiny, it is still significantly higher than that of YOLOv6n and YOLOv9-tiny, reaching 115.3 f/s. The improved YOLOv10n-BC model not only maintains high detection accuracy but also meets the real-time requirements of practical applications. This demonstrates that the innovative modules and structural optimizations proposed in this study significantly enhance the model's detection capabilities in complex driving scenarios, showcasing its superiority and potential in the task of distracted driving behavior detection. Since the CSAM [29] and CASSA modules have similar structures, this study conducted comparative experiments by replacing the CASSA module with CSAM in YOLOv10n. Both modules have a comparable number of parameters, with the CASSA module exhibiting slightly faster inference speed. Additionally, the CASSA module demonstrated better performance on the CBTDDD dataset.

5. Conclusion

This study developed a distracted driving behavior dataset that more accurately reflects real-world driving conditions by utilizing real-time surveillance images from various types of vehicles. By integrating the BiFPN and CASSA modules, the YOLOv10n model was enhanced, significantly improving the accuracy and efficiency of driver behavior detection. The enhanced model outperformed the original YOLOv10n and other mainstream object detection models across multiple evaluation metrics, particularly demonstrating a 2% increase in mAP@0.5, which signifies substantial progress in detection accuracy. Additionally, the model achieved excellent real-time detection performance with an FPS of 115.3 f/s, thereby fully meeting the requirements for real-

time monitoring applications. However, despite the notable increase in precision, the improvement in recall was relatively limited. This may result from the model's optimization process prioritizing the reduction of false positives, making it more conservative in decision-making and leading to some true positive instances going undetected. Future research will focus on optimizing the model's balancing strategies to further enhance recall while maintaining high precision, thereby achieving more comprehensive and balanced detection performance.

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